

Dataset Analysis for Decision-Making

March 19, 2025

 **sicor_municipal_co... .csv**
16.7 MB

Analyze the provided dataset to identify its key characteristics for data-driven decision-making. Upon receiving a specific analytical question, address it strictly using the dataset. Your deliverables must include a clear, concise answer to the question and the Python code that extracts and processes the data used to derive your answer.



qwen-max-latest

To analyze the dataset and address specific analytical questions, I will first load the dataset and examine its structure, key characteristics, and contents. This will allow me to understand the data's potential for data-driven decision-making. Once the dataset is understood, I will address the specific analytical question using Python code.

Step 1: Load and Inspect the Dataset

I will begin by loading the dataset (`sicor_municipal_contracts.csv`) into a pandas DataFrame and inspecting its structure.

```
python
1 import pandas as pd
2
3 # Load the dataset
4 file_path = 'sicor_municipal_contracts.csv'
5 df = pd.read_csv(file_path)
6
7 # Display basic information about the dataset
8 print("Dataset Overview:")
9 print(df.info())
10
11 # Display the first few rows of the dataset
12 print("\nFirst 5 Rows of the Dataset:")
13 print(df.head())
14
15 # Display summary statistics for numerical columns
16 print("\nSummary Statistics:")
17 print(df.describe())
```

Key Characteristics to Identify

- Columns and Data Types** : What are the column names, and what data types do they represent (e.g., numeric, categorical, datetime)?
- Missing Values** : Are there any missing or null values in the dataset?
- Unique Values** : How many unique values exist in categorical or identifier columns?
- Numerical Ranges** : What are the ranges of numerical columns (e.g., minimum, maximum, mean)?
- Potential Relationships** : Are there any obvious relationships or patterns between columns?

Step 2: Address the Analytical Question

Once the dataset is understood, I will address the specific analytical question. Please provide the analytical question you would like answered. For example:

- What is the total value of contracts awarded in each municipality?
- Which contractor has received the highest total contract value?
- How many contracts were awarded in each year?

After receiving the question, I will write Python code to extract and process the relevant data to derive a clear and concise answer.

Please provide the analytical question, and I will proceed with the analysis.





qwen-max-latest

To answer the question, "What is the number of different programs per state?", we will analyze the dataset to determine how many unique programs are associated with each state.

Approach:

- 1. **Identify Relevant Columns** : We need two columns: one for "state" and one for "program". The "state" column identifies the state, and the "program" column contains the names of the programs.
- 2. **Group by State** : For each state, we will identify the unique programs using the `nunique()` function.
- 3. **Output Results** : Display the count of unique programs for each state.

Here is the Python code to perform this analysis:

```
python
1 import pandas as pd
2
3 # Load the dataset
4 file_path = 'sicor_municipal_contracts.csv'
5 df = pd.read_csv(file_path)
6
7 # Ensure the necessary columns exist
8 if 'state' in df.columns and 'program' in df.columns:
9     # Group by 'state' and count unique programs
10    program_counts_per_state = df.groupby('state')['program'].nunique().reset_index()
11    program_counts_per_state.columns = ['State', 'Number of Unique Programs']
12
13    # Display the results
14    print("Number of Different Programs per State:")
15    print(program_counts_per_state)
16 else:
17    print("Error: Required columns ('state' or 'program') are missing from the dataset.")
```

Explanation of the Code:

- 1. **Loading the Dataset** : The dataset is loaded into a pandas DataFrame.
- 2. **Validation** : We check if the required columns (`state` and `program`) exist in the dataset.
- 3. **Grouping and Counting** : Using `groupby('state')` , we group the data by state, and then use `nunique()` on the `program` column to count the number of unique programs for each state.
- 4. **Formatting Output** : The result is formatted into a DataFrame with clear column names (`State` and `Number of Unique Programs`).
- 5. **Display Results** : The final table shows the number of unique programs for each state.

Example Output:

If the dataset contains the following data:

STATE	PROGRAM
Texas	Education
Texas	Healthcare
Florida	Infrastructure
Florida	Education
Florida	Education

The output would be:

STATE	NUMBER OF UNIQUE PROGRAMS
Texas	2
Florida	2

Next Steps:



Run the code above on the dataset. If you encounter any issues (e.g., missing columns or unexpected data formats), please provide additional details so I can refine the solution.

Let me know if you need further assistance!

